TRC-TMP1 Digital Temperature Sensor with I2C Interface

This is a fictional part, made up for the trace sample project. It is not a real product and no manufacturer publishes it. Do not order it.

1 Description

A +/-0.5 C temperature sensor with a 13-bit result (1/128 C per LSB in a 16-bit register, two's complement, left-justified), on a 400 kHz I2C bus at address 0x48 to 0x4B. The temperature register is register 0x00: write its address, then read two bytes, MSB first. 25.0 C reads 0x0C80.

Pin	Name	Description
1	SDA	I2C data
2	GND	Ground
3	A0	Address select
4	SCL	I2C clock
5	VDD	Supply, 1.8 to 3.6 V

7.5 I2C Timing Requirements

VDD = 3.3 V, TA = -40 C to 85 C, fast mode.

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
SCL clock frequency	fSCL				400	kHz
Hold time (repeated) START condition	tHD;STA		0.6			µs
Low period of the SCL clock	tLOW		1.3			µs
High period of the SCL clock	tHIGH		0.6			µs
Set-up time for a repeated START condition	tSU;STA		0.6			µs
Data hold time	tHD;DAT		0		0.9	µs
Data set-up time	tSU;DAT		100			ns
Set-up time for STOP condition	tSU;STO		0.6			µs
Bus free time between a STOP and START condition	tBUF		1.3			µs
Capacitive load for each bus line	Cb				400	pF

7.6 Electrical Characteristics

Parameter	Symbol	Conditions	Min	Typ	Max	Unit
Supply current	IDD	converting		45	75	µA
Temperature accuracy	TERR	0 to 70 C	-0.5		0.5	°C
Conversion time	tCONV			26	35	ms